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Your grade: 100%

Your best: 100%

Your highest score: 100%

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Module 1 Summative Assessment

Review Learning Objectives

Instructions

1. Which data type is associated with specific geographic locations?

Text Data

Image and Video Data

Panel Data

Geospatial Data

Correct

Due

Attempts

Correct: Geospatial data is associated with specific geographic locations.

Submitted Nov 14, 2024 1:02 PM CST

Try again

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice

Let's chat

1 / 1 point

2. Which of the following data types consist of measurable quantities?

Categorical Data

Image and Video Data

Numerical Data

Text Data

Correct

Like

Dislike

Report as issue

Your grade

To pass you need at least 80%. We keep your highest score.

100%

View submission

See feedback

1 / 1 point

3. If $L(Y_i, f(X_i))$ must be non-negative and continuous, which one of the following choices could be a loss function?

Median Absolute Error, $L(Y_i, f(X_i)) = \text{median}(|y_i - f(x_i)|)$

Mean Absolute Error, $L(Y_i, f(X_i)) = \frac{1}{2} \sum_{i=1}^n |y_i - f(x_i)|$

Maximum predicted value, $L(Y_i, f(X_i)) = \max_i(|y_i - f(x_i)|)$

Mode of Differences, $L(Y_i, f(X_i)) = \text{mode}(|y_i - f(x_i)|)$

Correct

1 / 1 point

4. What is a variable in the context of statistical learning?

A constant that remains unchanged.

A characteristic or attribute that can change or take on different values.

A specific outcome or result.

A type of statistical learning model.

Correct

1 / 1 point

5. Which data type tracks the same subjects over multiple time points?

Panel Data

Longitudinal Data

Time-series Data

Cross-sectional Data

Correct

1 / 1 point

6. Why is it important to place restrictions on what the function f can be?

When f has no restrictions, it must be the function that contains all the points in the dataset, which would be overfitting the data.

Limiting f ensures higher prediction accuracy on new data.

Restrictions on f enhance the computational speed of model training.

Restrictions make f more interpretable.

Correct

1 / 1 point

7. What does panel data combine?

Cross-sectional and Longitudinal Data

Numerical and Categorical Data

Geospatial and Image Data

Time-series and Cross-sectional Data

Correct

1 / 1 point

8. Why should one study statistical learning?

To build predictive models.

To create large datasets.

To understand data visualization.

To become a statistician.

Correct

1 / 1 point

9. Which statement best describes the difference between independent variables and features?

Features are obtained through experimental contexts, while independent variables may or may not be.

Independent variables and features always refer to the same concept.

Features are dependent variables.

Independent variables are assumed to be obtained through experimental contexts, while features may or may not be.

Correct

1 / 1 point

10. Which term is often synonymous with independent variables in the context of statistical learning?

Regression

Supervised Learning

Feature

Dependent Variable

Correct

1 / 1 point

11. What does discrete data represent in numerical data?

Data associated with specific geographic locations.

Countable values with gaps between them; often whole numbers.

Data collected at regular time intervals.

Values that can be any real number within a range.

Correct

1 / 1 point

12. What is the primary goal of statistical learning?

To create massive datasets.

To eliminate noise from data.

To make accurate predictions and draw meaningful conclusions from data.

To make datasets more complex.

Correct

1 / 1 point

13. Which type of data could be described as "higher dimensional"?

Categorical Data

Longitudinal data

Binary Data

Panel Data

Correct

1 / 1 point

14. True or False: The independent variable X_i in the formal description of the statistical learning problem is one-dimensional.

False

True

Correct

1 / 1 point

15. How is a dependent variable sometimes referred to?

Feature

Supervised Variable

Classification

Output Variable

Correct

1 / 1 point